Beyond Take-Over Request: Exploring Human Intervention and User Interface Design for Remote Assistance in Automated Vehicles

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Abstract

Remote assistance is emerging as an important approach for supporting automated vehicles in complex, ambiguous, or unforeseen situations. While remote assistance differs from traditional take-over in terms of operator location, control authority, intervention level, and scalability, both concern how humans should be involved when automated driving systems. Building on the well-established body of take-over request (TOR) research, this workshop examines how human intervention can be conceptualised and structured in remote assistance contexts. Participants will compare TOR and remote assistance, identify key dimensions of intervention, and discuss how these dimensions translate into user interface design implications. The workshop will produce an initial framework, interface design considerations, and a research agenda for supporting effective, timely, and scalable human intervention in remote assistance.

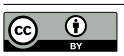
CCS Concepts

• **Human-centered computing → Interaction design**.

Keywords

Automated Driving, Remote Assistance, Human-Machine Interaction, Human Factors, Interaction Design

ACM Reference Format:
Soyeon Kim, Maytheewat Aramrattana, Jonas Andersson, and Michael Hildebrandt. 2026. Beyond Take-Over Request: Exploring Human Intervention and User Interface Design for Remote Assistance in Automated Vehicles. In *18th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI Adjunct '26), September 20–23, 2026, Gothenburg, Sweden.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3828158.3834792



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AutomotiveUI Adjunct '26, Gothenburg, Sweden
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ACM ISBN 979-8-4007-2815-0/26/09
https://doi.org/10.1145/3828158.3834792

1 Introduction

Despite significant advances in automated driving technologies, human involvement remains necessary for real-world deployment. In complex, ambiguous or unexpected situations, automated systems alone may be insufficient and human judgment and intervention may still be required. Therefore, remote assistance has emerged and is already being deployed as a new approach to ensuring system safety and reliability [14]. Compared to remote driving, in which remote operators fully perform dynamic driving tasks such as acceleration, braking, and steering, remote assistance is characterised by event-driven, rather than continuous, intervention [1]. In remote assistance, the driving automation system retains longitudinal and lateral control, while human remote assistance operators provide high-level judgments when the system encounters uncertainty or lacks contextual understanding. Such cases include verifying the legality of unusual detours, interpreting ambiguous traffic signs, or confirming the permissibility of specific manoeuvres. These functions remain relevant even in highly automated deployments, particularly Level 4 systems operating within defined operational design domains. For example, Waymo operates a "fleet response" system that provides guidance when vehicles face unexpected situations, although it does not directly control the vehicles [18].

Traditionally, one important concept in human involvement with automated vehicles has been the take-over request (TOR). TOR describes the transfer of control from an automated system to a human driver. Over the past decade, TOR has been extensively studied, providing a structured understanding of human–automation interaction in terms of timing [8], take-over time [19], and user interface [10] support. This body of work has established a strong foundation for designing and evaluating human–automation interactions in automated vehicles.

Remote assistance and TOR share a conceptual similarity: both concern how humans should be involved when automated driving systems encounter uncertainty or reach the limits of their capabilities. In both cases, the automated system must communicate its need for human input, support human situation assessment, and enable timely intervention. This similarity is also reflected in user interface research. TOR studies have examined how UI can

direct drivers' attention to relevant hazards, explain automation and traffic context, and help drivers anticipate upcoming events. Similar concerns arise in remote-operation UI studies, even when the operator is physically separated from the vehicle. For example, in TOR contexts, AR-HUDs have been used to highlight takeover-relevant cues for in-vehicle drivers [12, 13], while remote operation studies have also presented the reasons for intervention to remote operators [4, 17]. In addition, predictive HMIs in TOR studies have supported drivers' safe driving performance after transitions [3, 5], whereas remote operation studies have employed predictive driving corridors, trajectory guidance, and related visualisations to help remote operators anticipate vehicle motion and the potential consequences of possible interventions [6, 9, 16]. Therefore, although TOR and remote operation differ in physical setting and control authority, UI research in both domains shares a common design concern: how user interfaces can help users understand why intervention is needed and act appropriately within a limited time window. However, remote assistance also differs fundamentally from TOR. Whereas TOR typically involves a transfer of vehicle control from automation to an in-vehicle driver, remote assistance often involves partial, high-level, or advisory intervention by an operator who is physically separated from the vehicle. As a result, remote assistance research has largely focused on helping operators gain situation awareness of the environment in which intervention is required [15]. For example, prior studies have designed interfaces that provide alternative vision perspectives, such as bird's-eye views [2], or that support immersive perception through audio [11], and haptic [7] feedback so that the remote environment can be experienced more like an actual vehicle context. In this sense, remote assistance differs from TOR in the nature of human–automation interaction and raises several important gaps in existing research.

We identified some of the research gaps as follows. First, existing work on in-vehicle take-over is centred on full control transfer and does not sufficiently account for partial or high-level intervention. Second, the binary framing of control authority, human versus automation, limits the ability to describe shared or distributed decision-making and execution. Third, situation awareness models based on in-vehicle drivers do not fully capture the mediated perception experienced by remote operators. Fourth, the focus on single transition events overlooks the broader intervention lifecycle, including detection, decision-making, action, and recovery. Finally, take-over is generally premised on an interaction between a single vehicle and a single human driver, whereas remote assistance is performed within the organisational and operational context of a remote operation centre. This raises two design challenges. First, remote assistance may need to be understood not as a standalone function, but as one integrated with monitoring and supervision tasks. This raises the question of whether remote assistance and remote monitoring should be supported by a unified user interface or by separate interfaces tailored to their distinct task characteristics. Second, remote assistance introduces new scalability challenges, as a single operator may supervise multiple vehicles, raising issues of attention allocation and prioritisation.

For these reasons, the TOR literature provides a useful but incomplete conceptual foundation for examining human–machine interaction in remote assistance. It offers mature concepts, such as control authority, transition timing, human readiness, situation awareness, and interface support, that can be re-examined in a remote and partially mediated context. At the same time, comparing remote assistance with TOR helps reveal where new concepts are needed: the human may not take over control, may intervene only at a strategic or tactical level, and may support multiple vehicles through mediated information.

This workshop, therefore, uses TOR as a conceptual reference point to systematically explore how human involvement is extended in remote-assistance contexts. We introduce the concept of human intervention as a unifying lens to capture diverse forms of involvement beyond full control transfer. Building on this, we aim to identify key dimensions, such as control level, timing, interaction granularity, information mediation, and operator–vehicle allocation, that structure intervention and inform interface design. Ultimately, the workshop seeks to provide a more comprehensive understanding of human–automation interaction in remote assistance and to explore interaction design directions that support effective, timely, and scalable human intervention.

2 Workshop Goals

The goal of this workshop is to develop a structured understanding of human intervention in remote assistance systems and to explore its implications for interface design. Specifically, the workshop examines how human involvement in remote assistance extends, modifies, and challenges assumptions established in take-over request (TOR) research. It also aims to identify interface design considerations that can support different forms of intervention in remote assistance. Through this process, the workshop seeks to outline a structured design space and research agenda for future work on human–automation interaction in remote assistance. The workshop is guided by the following research questions:

- How can human intervention in remote assistance be conceptualised beyond traditional take-over?

- What are the key dimensions that structure human intervention in remote assistance?

- How can these dimensions be reflected in the design of remote assistance user interfaces?

- How can remote assistance interfaces support scalable intervention across multiple vehicles while integrating or separating monitoring and assistance functions?

To address these questions, the workshop will explore the following directions: 1. Forms of intervention: examining different types of human involvement beyond full control transfer, such as judgment, confirmation, recommendation, and strategic guidance. 2. Dimensions of intervention: identifying how intervention can be structured in terms of control level, timing, authority, interaction granularity, and information mediation. 3. User interface design implications: discussing how remote assistance interfaces can support situation assessment, timely decision-making, and effective communication between operators and vehicles. 4. Scalability of intervention: considering how interfaces can support scalable intervention when one operator supervises multiple vehicles, including attention allocation and prioritisation, as well as the integration or separation of monitoring and assistance functions.

3 Expected Outcomes

The workshop will produce a shared conceptual and design-oriented understanding of human intervention in remote assistance systems. Its main outcome will be an initial framework that clarifies how concepts from takeover request research can be extended, modified, or challenged in remote assistance contexts. Based on this framework, the workshop will identify key dimensions of intervention, such as control level, timing, authority, information mediation, and operator-vehicle allocation and translate them into user interface design implications and an initial design space for remote assistance. After the workshop, the results and insights will be systematically documented and shared with participants. We also plan to establish a communication channel to support continued collaboration, which may lead to joint research initiatives, follow-up workshops, collaborative publications, or dissemination through a joint paper or special issue.

4 Workshop Schedule

The workshop is designed as a 3-4 hour half-day session, including two short breaks. It consists of three main phases following the opening session.

· **Introduction and Ignition talks** (30 minutes): The workshop will start with the organizers and participants introduction. Short ignition talks introduce key perspectives on TOR, remote assistance, and human intervention. A brief interactive discussion follows to establish a shared vocabulary and identify initial assumptions and open questions.

· **Group interactive session 1** (40 minutes) - Comparative mapping from TOR to remote assistance: Participants examine how key TOR concepts, such as control transfer, readiness, situation awareness, timing, and interface support, apply to remote assistance. The aim is to identify which concepts can be transferred, which need to be modified, and which are challenged in remote assistance contexts. In addition, participants are invited to offer conceptual, technical, and practical perspectives that may broaden the problem space beyond the initial TOR-based framing.

· **Break** (15 minutes)

· **Group interactive session 2** - Structuring human intervention (40 minutes): Building on the comparative mapping, participants derive key dimensions for describing human intervention in remote assistance. To help ground the discussion in concrete scenarios, participants work with remote assistance use cases, such as manoeuvre confirmation, object identification, and tactical manoeuvre suggestion. Participants also consider how these intervention cases are embedded within broader monitoring and supervision workflows in a remote operation centre, rather than treating remote assistance as a standalone function. These cases serve as prompts for examining how intervention may differ in terms of control level, timing, authority, information mediation, interaction granularity, and operator–vehicle allocation.

· **Group interactive session 3** - Interface implications (40 minutes): Participants discuss how the intervention dimensions identified in Phase 2 translate into requirements for the design of remote assistance user interfaces. The discussion is grounded in the use cases examined in interactive session 2, allowing participants to consider how different forms of remote assistance may require varying levels of user interface support. The discussion focuses on situation assessment, decision support, and communication between operators and vehicles, while also addressing interface support for multi-vehicle supervision, attention allocation, urgency assessment, and prioritisation of assistance requests.

· **Break** (15 minutes)

· **Wrap-up session** (30 minutes): The workshop concludes with a collective synthesis of the main insights from the previous sessions. Participants identify recurring themes, unresolved challenges, and key directions for future research. The session also outlines how the workshop outcomes will be documented, shared with participants, and potentially developed into follow-up collaborations, joint publications, or future workshops.

5 Attendance

The workshop is designed for approximately 15-20 participants from both academia and industry, including researchers and practitioners in automotive user interfaces, human–computer interaction, human factors, and remote operation systems. We particularly welcome participants with an interest in user interface design and human involvement in automated driving. The workshop will emphasise active participation and collaboration, encouraging participants to exchange perspectives and co-develop conceptual frameworks.

6 Biographies

Dr Soyeon Kim is an assistant professor at University of Warwick. Her research focuses on understanding human behaviour and designing human-machine interaction in automated vehicles. She holds a PhD at TU Delft in designing in-vehicle HMI for trustworthy automated vehicles.

Dr Maytheewat Aramrattana is a senior researcher at VTI. His research focuses on remote operation of road vehicles, and simulation-based testing and evaluation of cooperative intelligent transport systems. He holds a PhD in Computer Science and Engineering from Halmstad University.

Dr Jonas Andersson is a principal scientist at the Institute for Energy Technology in Halden, Norway. His research focus on control room operator work and interaction design in safety critical industries with a special interest in Human-AI collaboration. He holds a PhD in Human-Technology-Design from Chalmers University of Technology in Gothenburg, Sweden.

Michael Hildebrandt is a senior scientist at the Institute for Energy Technology in Norway where he leads the Biometrics Lab and Human-AI Lab. His research focuses on digital control rooms, multi-agent collaboration, and user monitoring systems.

References

[1] Daniel Bogdoll, Stefan Orf, Lars Töttel, and J Marius Zöllner. 2022. Taxonomy and survey on remote human input systems for driving automation systems. In *Future of Information and Communication Conference*. Springer, 94–108.

[2] A. Boker and Joel Lanir. 2023. Bird's eye view effect on situational awareness in remote driving. In *Adjunct Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.

[3] Guy Cohen-Lazry, N. Katzman, Avinoam Borowsky, and Tal Oron-Gilad. 2019. Directional tactile alerts for take-over requests in highly-automated driving. *Transportation Research Part F: Traffic Psychology and Behaviour* 65 (2019), 217–226.

[4] Mark Colley, J. Westhauser, Jonas Andersson, Alexander G. Mirnig, and Enrico Rukzio. 2025. Introducing ROADS: A systematic comparison of remote control

interaction concepts for automated vehicles at road works. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*.

[5] Alexander Eriksson, Sebastiaan M. Petermeijer, Markus Zimmermann, Joost C. F. de Winter, Klaus J. Bengler, and Neville A. Stanton. 2019. Rolling out the red and green carpet: Supporting driver decision making in automation-to-manual transitions. *IEEE Transactions on Human-Machine Systems* 49, 1 (2019), 20–31.

[6] Michael Gafert, Alexander G. Mirnig, Peter Fr"ohlich, Verena Kraut, Z. Anzur, and Manfred Tscheligi. 2023. Effective remote automated vehicle operation: A mixed reality contextual comparison study. *Personal and Ubiquitous Computing* 27, 6 (2023), 2321–2338.

[7] J. M. Georg and Frank Diermeyer. 2019. An adaptable and immersive real time interface for resolving system limitations of automated vehicles with teleoperation. In *Proceedings of the IEEE International Conference on Systems, Man and Cybernetics*.

[8] Christian Gold, Daniel Damböck, Lutz Lorenz, and Klaus Bengler. 2013. "Take over!" How long does it take to get the driver back into the loop?. In *Proceedings of the human factors and ergonomics society annual meeting*, Vol. 57. Sage Publications Sage CA: Los Angeles, CA, 1938–1942.

[9] Gregor Graf and Heinrich Hussmann. 2020. User requirements for remote teleoperation-based interfaces. In *Proceedings of the 12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.

[10] Soyeon Kim, René van Egmond, and Riender Happee. 2021. Effects of user interfaces on take-over performance: a review of the empirical evidence. *Information* 12, 4 (2021), 162.

[11] Pontus Larsson, J. B. R. de Souza, and Jakob Begnert. 2023. An auditory display for remote road vehicle operation that increases awareness and presence. In *Proceedings of the 28th International Conference on Auditory Display*. Norrk"oping, Sweden.

[12] Patrick Lindemann, N. M"uller, and Gerhard Rigolll. 2019. Exploring the use of augmented reality interfaces for driver assistance in short-notice takeovers. In *Proceedings of the IEEE Intelligent Vehicles Symposium*.

[13] L. Lorenz, Philipp Kerschbaum, and J. Schumann. 2014. Designing take over scenarios for automated driving: How does augmented reality support the driver to get back into the loop?. In *Proceedings of the Human Factors and Ergonomics Society 58th Annual Meeting*.

[14] Edward J. Markey. 2026. Remote Backseat Operators: Revealing the Autonomous Vehicle Industry's Reliance on Human Remote Assistance Operators. Accessed: 2026-06-15. https://www.markey.senate.gov/imo/media/doc/remote_assistance_investigation_report.pdf

[15] Andreas Schrank, Fabian Walocha, Stefan Brandenburg, and Michael Oehl. 2024. Human-centered design and evaluation of a workplace for the remote assistance of highly automated vehicles. *Cognition, Technology & Work* 26, 2 (6 2024), 183–206. doi:10.1007/s10111-024-00753-x

[16] Yuji Tazaki, R. Marumoto, J. Ohgashira, H. Nagano, Y. Yokokohji, and S. Kameoka. 2024. Experimental evaluation of the effect of predictive display on steering stability in remote driving. *International Journal of Automotive Engineering* 15, 2 (2024), 90–97.

[17] Felix Tener and Joel Lanir. 2022. Driving from a distance: Challenges and guidelines for autonomous vehicle teleoperation interfaces. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*.

[18] Waymo. 2024. Fleet response: Lending a helpful hand to Waymo's autonomously driven vehicles. Accessed: 2026-05-27. https://waymo.com/blog/2024/05/fleet-response

[19] Bo Zhang, Joost De Winter, Silvia Varotto, Riender Happee, and Marieke Martens. 2019. Determinants of take-over time from automated driving: A meta-analysis of 129 studies. *Transportation research part F: traffic psychology and behaviour* 64 (2019), 285–307.